The Shift — Musical System Map

Score documentation for a generative, interactive VR composition

Semiha Paksoy · MUS 442 Senior Project II · Sound Installation and Multimedia Project · June 2026

1. How to Read This Document

The Shift is composed as a set of generative, interactive musical systems rather than as a fixed score. The music is created at runtime by rule systems whose parameters are written in the project's code, and it is performed jointly by the systems and the player. Conventional staff notation can therefore capture the pitch material, but not the piece itself.

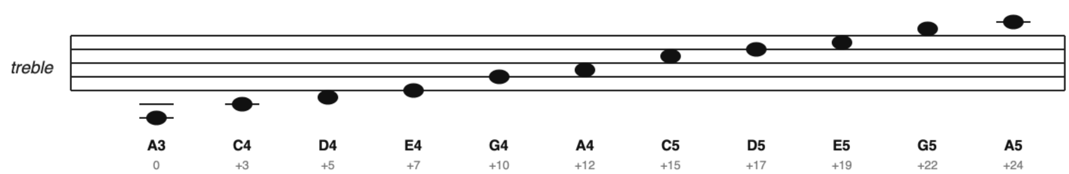
This document is the project's notation: a system map that specifies, for each scene, the pitch reservoir, the time structure, the sounding objects, and the rules that map player interaction to musical events. All numeric values (tempi, loop periods, pitch sets, step durations) are taken directly from the implemented Unity scripts — FactoryMusicDirector, StreetAmbienceDirector, MetroMusicDirector, ConcertAudioDirector, RadialSequencer, and SequencerFinaleDirector. A performer who followed these rules on acoustic or electronic instruments would produce a recognizable performance of the work; the Unity build realizes them exactly.

2. Shared Musical Language

All four scenes draw from a single pitch reservoir — the minor pentatonic scale rooted on A — and a single timing authority, Unity's sample-accurate DSP clock (AudioSettings.dspTime). Every scheduled event in the project is quantized to a scene-specific sixteenth-note grid on this clock.

Shared Pitch Reservoir — A Minor Pentatonic, A3 to A5

Semitones above A3 (220 Hz): 0 · 3 · 5 · 7 · 10 · 12 · 15 · 17 · 19 · 22 · 24 — used by the factory melody, metro ensemble, and sequencer orbs



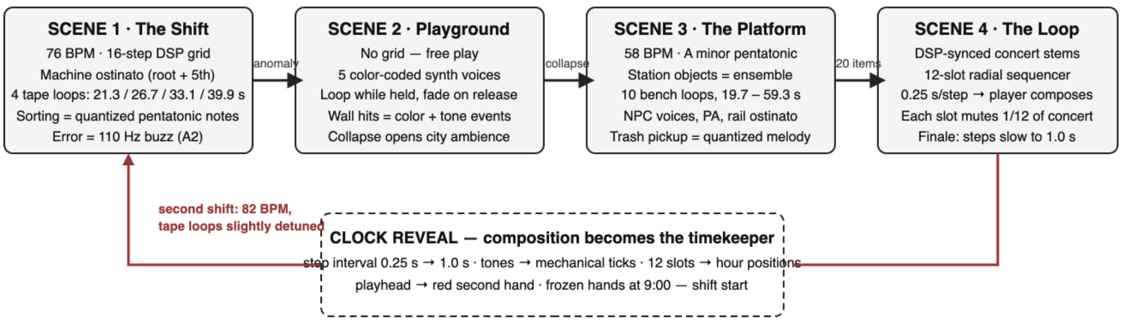
Scene 1 tape loops transpose the same set below the root: A (0), E (−5), C (+3), D (+5). The error buzz is A2 (110 Hz); the lamp drone stacks A2 + E3 + A3.

- Pentatonic degree set: {0, 3, 5, 7, 10} semitones, extended over two octaves.
- Reference pitches: A2 = 110 Hz (error buzz, drone root), A3 = 220 Hz (sequencer reference).
- Interaction events (sorting, trash deposits, orb placements) never sound immediately: they are scheduled on the next grid step, so the player is always in time.

3. The Four-Scene Arc

Each scene changes the player's musical role: executor of the factory's score, free improviser, participant inside an autonomous ensemble, and finally composer — whose finished composition reveals itself as the factory clock.

The Shift — Musical Arc of the Four-Scene Loop

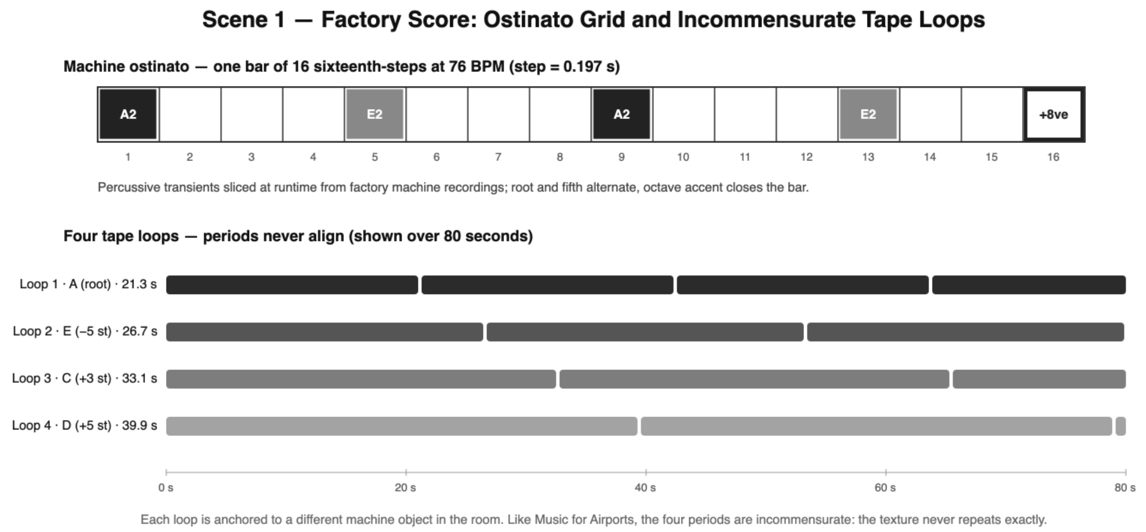


All scenes share one pitch language (minor pentatonic rooted on A) and sample-accurate AudioSettings.dspTime scheduling.

4. Scene 1 — The Shift (Factory)

The factory score is assembled at load time from machine recordings in Resources/scene_1_sound_design. The recordings are sliced into percussive transients, tonal loop material, and steam breaths; the slices are then scheduled as an ostinato plus four incommensurate tape loops in the manner of Brian Eno's Music for Airports.

Parameter	Value
Tempo	76 BPM (second shift: 82 BPM); sixteenth-step = 0.197 s
Ostinato	16-step grid: root on steps 1 and 9, low fifth on 5 and 13, octave accent at bar end
Tape loops	Periods 21.3 / 26.7 / 33.1 / 39.9 s at 0 / −5 / +3 / +5 semitones (A, E, C, D)
Sort feedback	Correct: next pentatonic melody note, register rises with quota. Wrong: error figure over a 110 Hz (A2) buzz
Second shift	Tempo +6 BPM; half of the tape loops slightly detuned
Mix	Ostinato 0.65 · tape loops 0.45 · steam 0.35 · melody notes 1.0 (shared RMS reference across scenes)



5. Scene 2 — The Colorful Playground

The playground abandons the grid: it is the only scene without a tempo. Twenty cubes (five colors × four) are the instruments. Each color is bound to one synth recording from Resources/scene_2_sound_design, loudness-normalized so no color dominates.

Rule	Behavior
Grab	The cube’s color loop starts from its body (3D source) and plays while held
Release	The loop fades out — sound is tied to touch, not possession
Throw / impact	Collision sound scaled by impact velocity; above 0.3 m/s the wall takes the cube’s color
Color voices	Red, Blue, Green, Yellow, Purple — one synth timbre each
Progression	20 total wall hits collapse the room; the enclosed synth space opens into procedural city ambience (StreetAmbienceDirector: rumble, car passes, intermittent voices, shop radio)

6. Scene 3 — The Platform (Metro)

The station is a distributed ensemble: every significant object class carries a musical layer, all locked to one 58 BPM DSP grid in A minor pentatonic, but each repeating at its own incommensurate period. The result is a texture that is always coherent and never identical — the musical image of the commuter cycle.

Scene 3 — The Station as Ensemble (58 BPM, A minor pentatonic)

Layer	Sound source	Time structure	Pitch content
1 · Rail ostinato	Track objects	16-step grid, sparse hits	Deep metallic A, low register
2 · Bench pads	10 Metalbench objects	Periods 19.7 → 59.3 s, free-running	Pentatonic dyads (table below)
3 · Lamp drone	Fluorescent lighting fixtures	Continuous hum	A2 + E3 + A3 (root, 5th, octave)
4 · Trash-can ticks	Each MetalTrashCan	Sparse 2-step pattern from index	Pentatonic, low register
5 · NPC voices	Each commuter body	Periods 11.3 / 13.7 / 17.9 / 19.3 / 23.7 / 29.3 s	Hum, murmur, whisper
6 · Tunnel rumble	Tunnel mouths	Continuous, distance-attenuated	Unpitched low noise
7 · PA announcements	Platform speakers	Triggered by train events + idle timer	TTS speech, ducked mix
8 · Trash melody	Player action (cart deposit)	Quantized to next 16th step	Rising pentatonic line, 2 octaves

Bench dyads (semitones above A3 → pitches):

{0, 7} = A3+E4 {3, 12} = C4+A4 {5, 12} = D4+A4 {7, 15} = E4+C5 {10, 19} = G4+E5

Every layer is scheduled on the same dspTime grid but repeats at its own incommensurate period — coherent, never identical, like the commuter cycle itself.

The player's contribution is the trash-collection melody: each item deposited in the cart plays the next note of a rising two-octave pentatonic line, quantized to the grid. Twenty items complete the scene.

7. Scene 4 — The Concert / The Loop

7.1 Concert Stems

ConcertAudioDirector starts all stems on a single dspTime timestamp. Some stems are 2D stereo (heard as a mix), others are 3D sources placed on stage objects (drum kit, instruments, speaker stacks). Stems are normalized to a common RMS target — the measured spread of the source material was 14 dB (drums −18.7, bass −24.3, FX −32.4 LUFS) — so per-stem volumes act as offsets above an equal floor.

7.2 The Radial Sequencer

The finale instrument is a twelve-slot ring — deliberately the geometry of a clock face. The player grabs sample orbs, previews them, and places them into slots. The DSP-locked playhead sounds each filled slot as it passes; each filled slot also mutes one twelfth of the concert, so completing the composition silences the band.

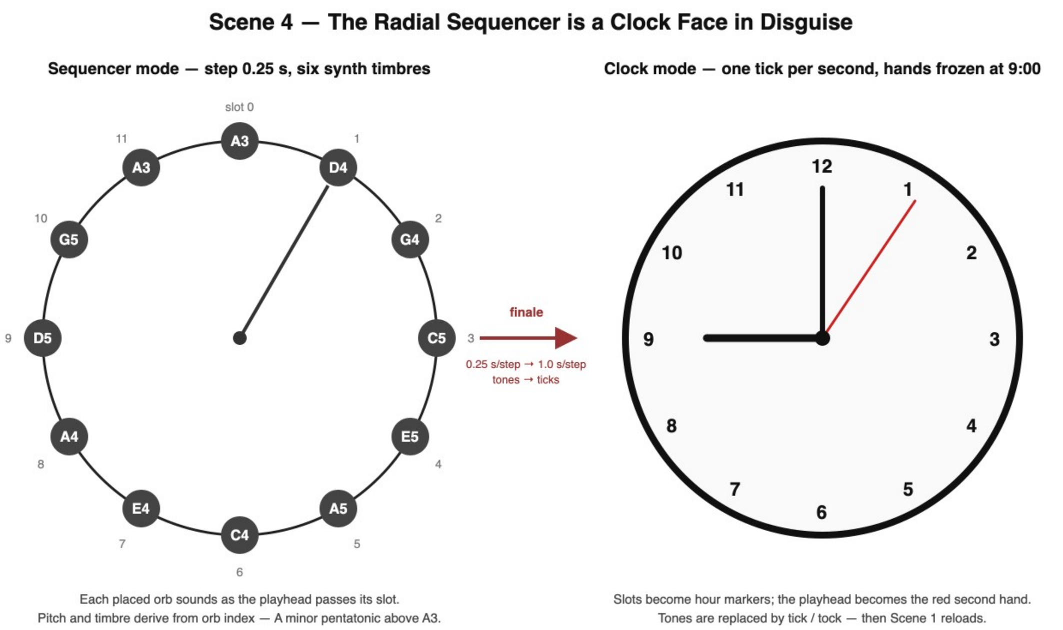
Orb #	Semitones above A3	Pitch	Timbre family
0	0	A3	1 of 6 synth timbres; timbre = orb index mod 6
1	5	D4	—
2	10	G4	—
3	15	C5	—
4	19	E5	—
5	24	A5	—
6	3	C4	—

7	7	E4	—
8	12	A4	—
9	17	D5	—
10	22	G5	—
11	0	A3	—

Step interval: 0.25 s (a playful 3-second loop). All orb tones are generated procedurally at load — the sequencer needs no audio assets.

7.3 The Clock Reveal

When all twelve slots are filled and the loop completes one full revolution, SequencerFinaleDirector takes over: the step interval slides from 0.25 s to exactly 1.0 s, tickDominance rises from 0 to 1 so tones become mechanical ticks, colors drain to the Scene 1 wall-clock palette, numbers 1–12 fade in, the playhead becomes a red second hand, and frozen hands appear at 9:00 — shift start. After six seconds of pure ticking, Scene 1 reloads at 82 BPM.



8. Interaction-to-Music Summary

Scene	Player gesture	Musical result
1 · Factory	Correct sort / wrong sort	Quantized rising pentatonic note / error figure on A2 buzz
2 · Playground	Grab, throw, wall hit	Color synth loop; velocity-scaled impact; wall repaint tone
3 · Metro	Deposit trash in cart	Next note of quantized pentatonic melody (20 notes)
4 · Concert	Place orb in slot	New voice in the loop; concert ducks

		by 1/12
4 · Finale	Complete the ring	Composition decelerates into the factory clock; ticks replace tones

Authorship: all musical systems, their parameters, and their narrative integration were designed and implemented by Semiha Paksoy. Source recordings (machine sounds, synth colors, TTS announcements, concert stems) are production materials shaped entirely by these systems.